

Course-End Project

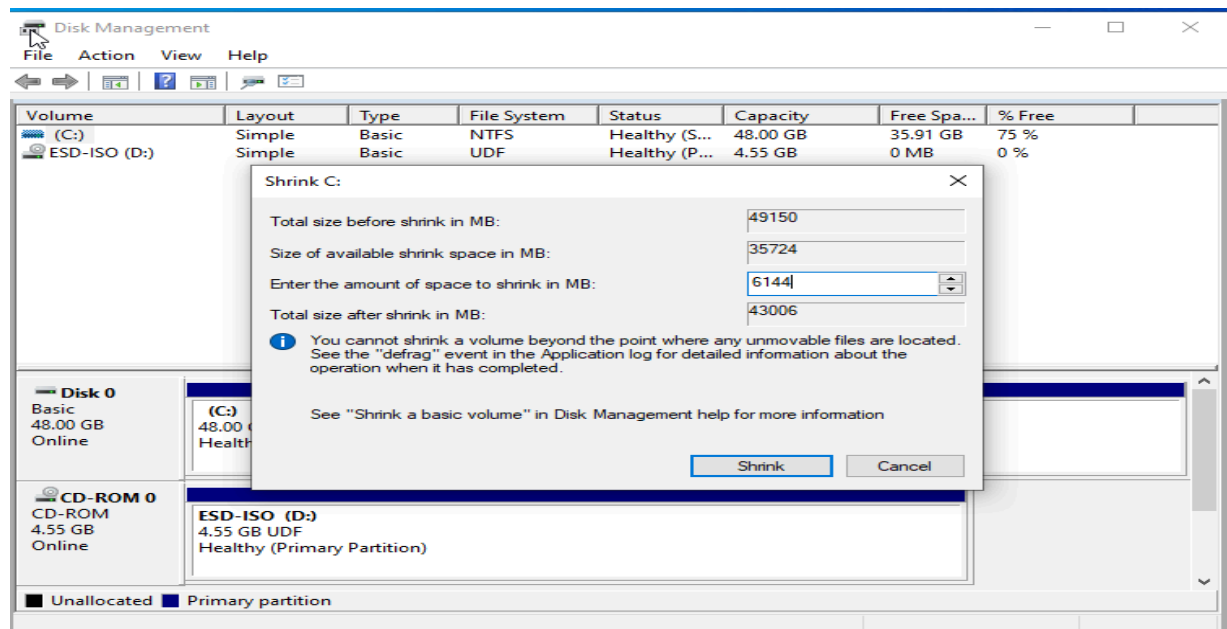
Digital Forensics and Steganography: Recovering Hidden Data from a JPG Image

Steps to the followed:

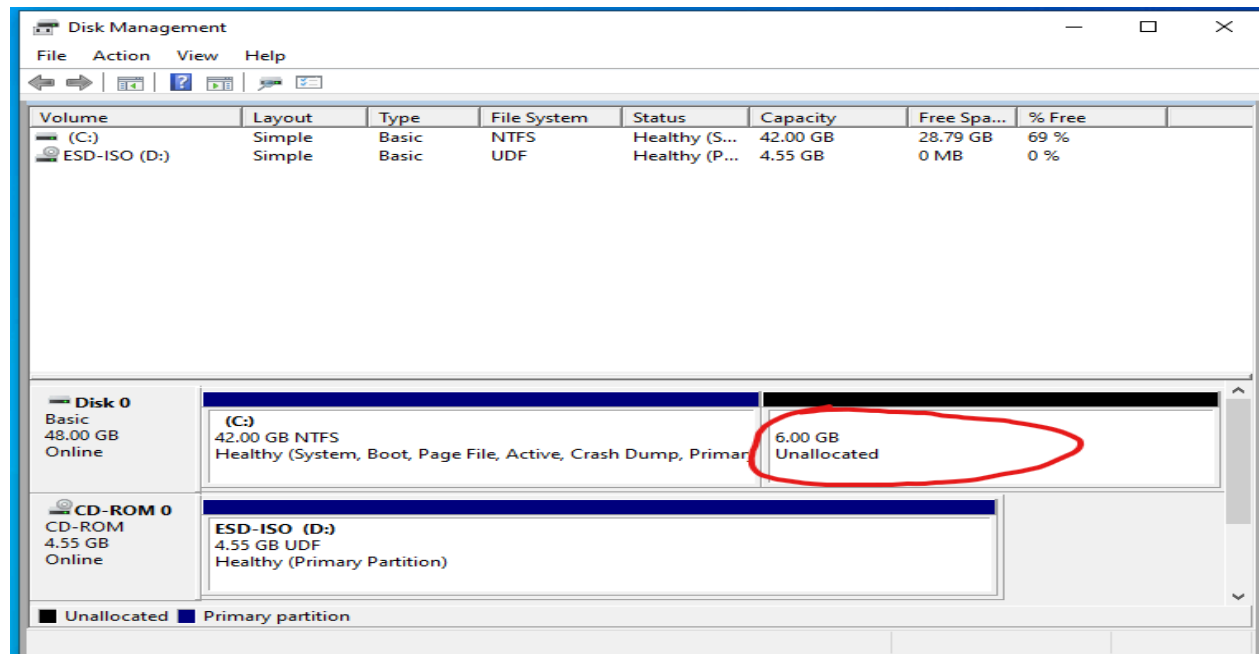
1. *Creating a 2-Disk System:* Provides a realistic scenario for recovering deleted files from storage environments.
2. *Prepare the JPG file for testing:* Create a standard JPG image and embed hidden text using a steganography tool(Silent Eye). Place the JPG image with hidden text onto a 2GB NTFS partition. Delete the JPG image to simulate file removal
3. *Recovery and decoding:* Recover the deleted image using FTK Manager and extract hidden text.

1. Creating a 2-Disk System

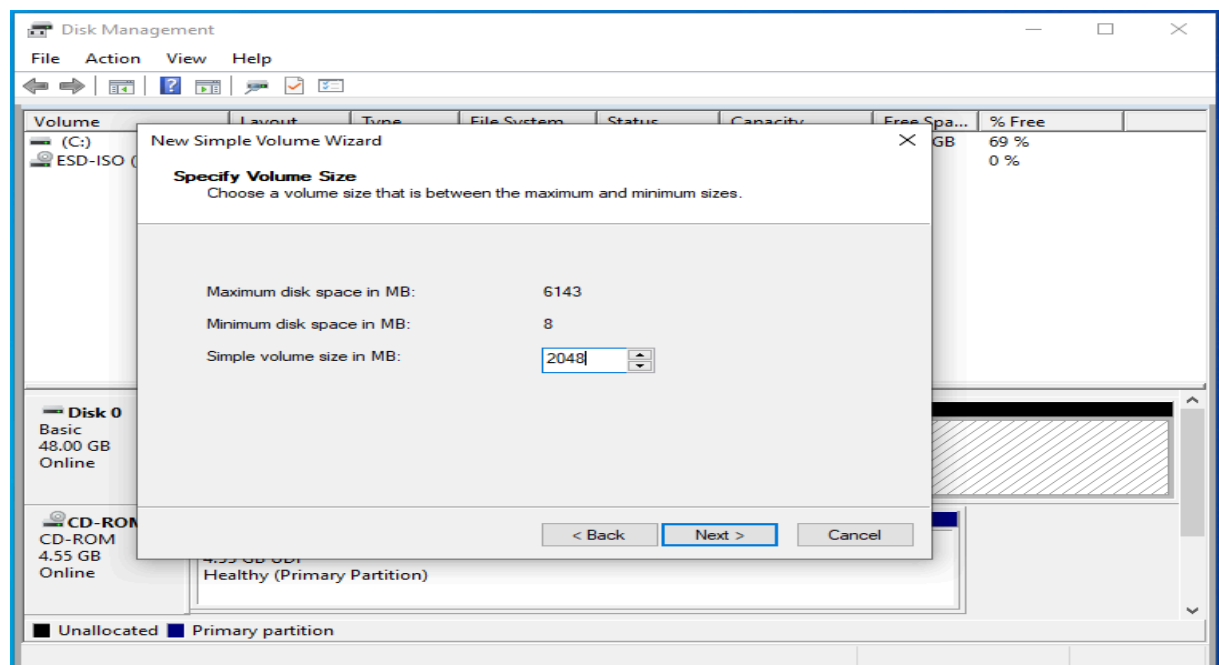
- 1.1. Open Windows VM.
- 1.2. In the search bar, type "Disk Manager" and open the Disk Manager console.
- 1.3. Right click on C: Drive and select *Shrink Volume*.
- 1.4. In the *Enter the amount of space to shrink* text box enter 6144 (6GB)



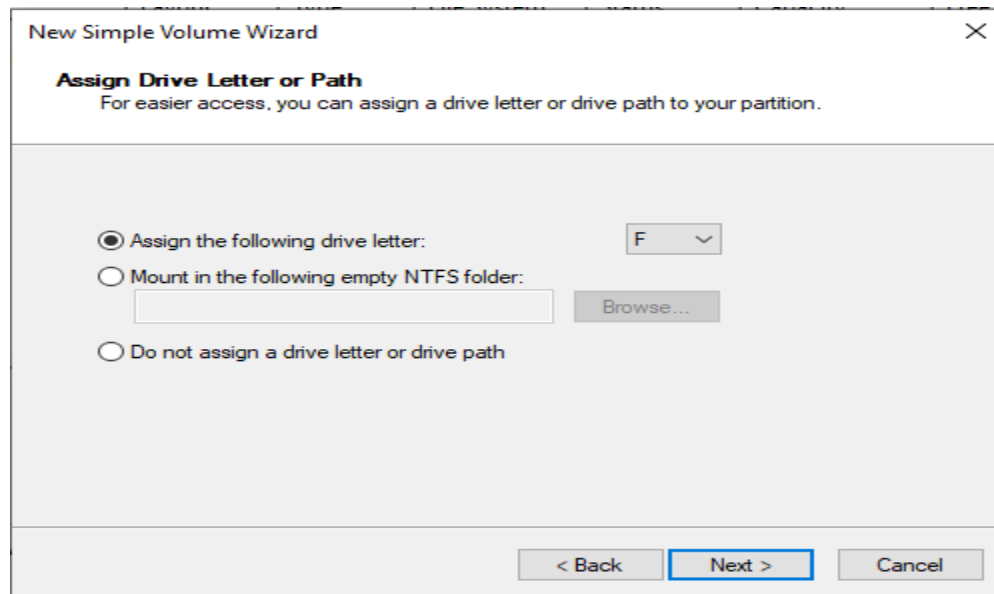
- 1.5. Click on the *Shrink* button. Back in the console you will see 6GB of unallocated memory.



- 1.6. Right click on the Unallocated space and select *New Simple Volume*. On the wizard window, click on Next.
- 1.7. Set *Simple volume size* as 2048, then click on Next.

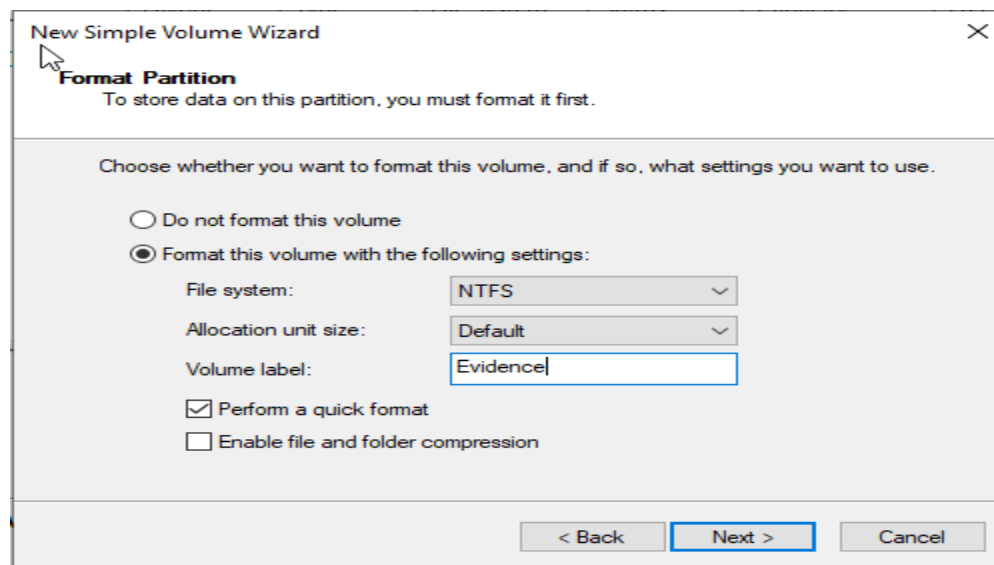


- 1.8. Assign F as the letter for the new drive and click on Next.



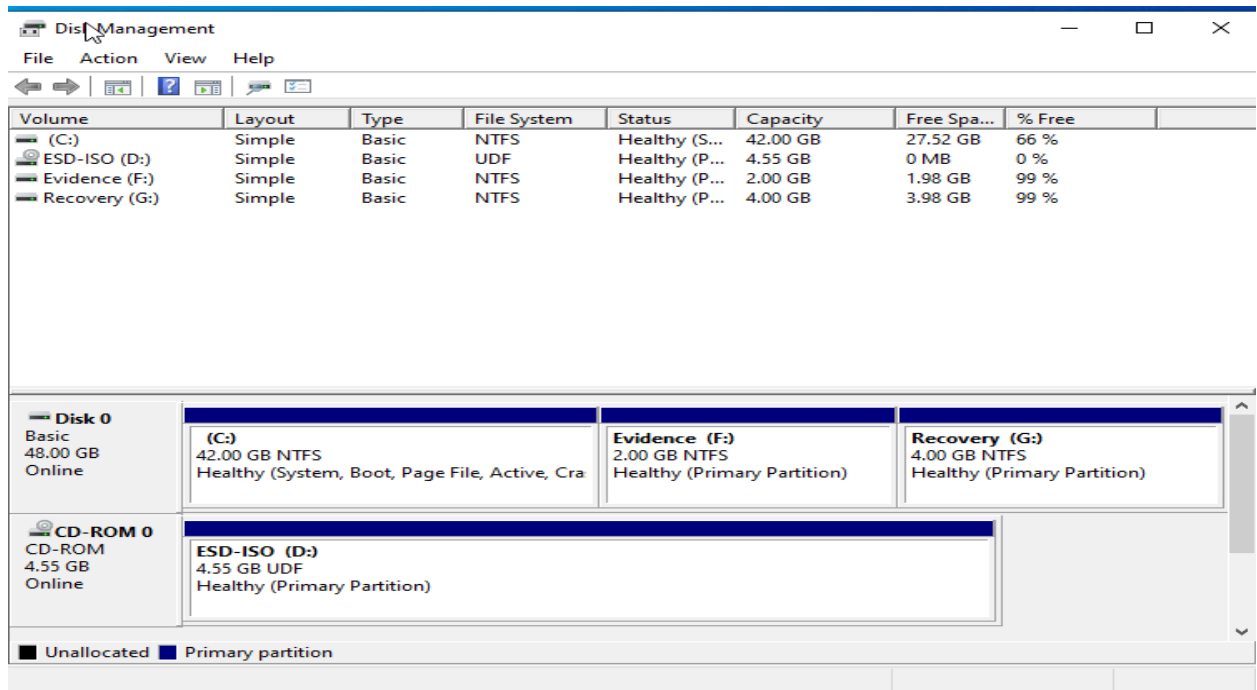
The screenshot shows the 'New Simple Volume Wizard' window at the 'Assign Drive Letter or Path' step. The window title is 'New Simple Volume Wizard' with a close button (X) in the top right corner. Below the title bar, the section is titled 'Assign Drive Letter or Path' with a subtitle 'For easier access, you can assign a drive letter or drive path to your partition.' There are three radio button options: 'Assign the following drive letter:' (selected), 'Mount in the following empty NTFS folder:', and 'Do not assign a drive letter or drive path'. The first option has a dropdown menu showing 'F'. The second option has a text box and a 'Browse...' button. At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

- 1.9. Type "Evidence" as the Volume label, then click on Next and then Finish.



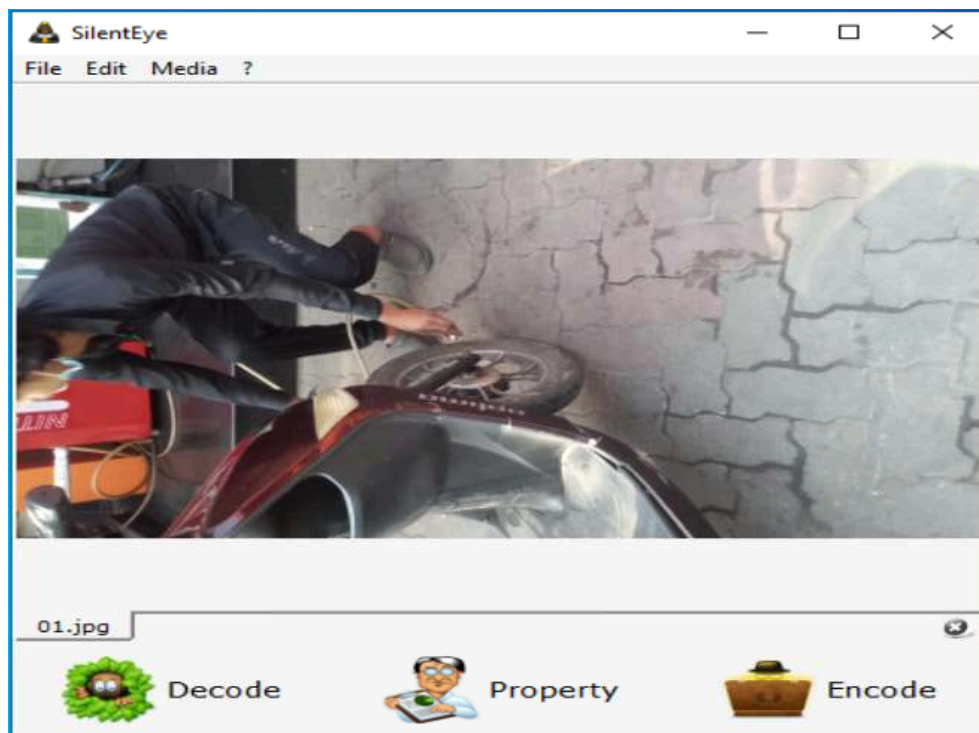
The screenshot shows the 'New Simple Volume Wizard' window at the 'Format Partition' step. The window title is 'New Simple Volume Wizard' with a close button (X) in the top right corner. Below the title bar, the section is titled 'Format Partition' with a subtitle 'To store data on this partition, you must format it first.' There is a text instruction: 'Choose whether you want to format this volume, and if so, what settings you want to use.' There are two radio button options: 'Do not format this volume' and 'Format this volume with the following settings:' (selected). Under the second option, there are three settings: 'File system:' (dropdown menu showing 'NTFS'), 'Allocation unit size:' (dropdown menu showing 'Default'), and 'Volume label:' (text box containing 'Evidence'). There are also two checkboxes: 'Perform a quick format' (checked) and 'Enable file and folder compression' (unchecked). At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

- 1.10. Perform the same steps for the remaining 4gb of unallocated memory, assign G as the volume letter and “Recovery” as volume label.

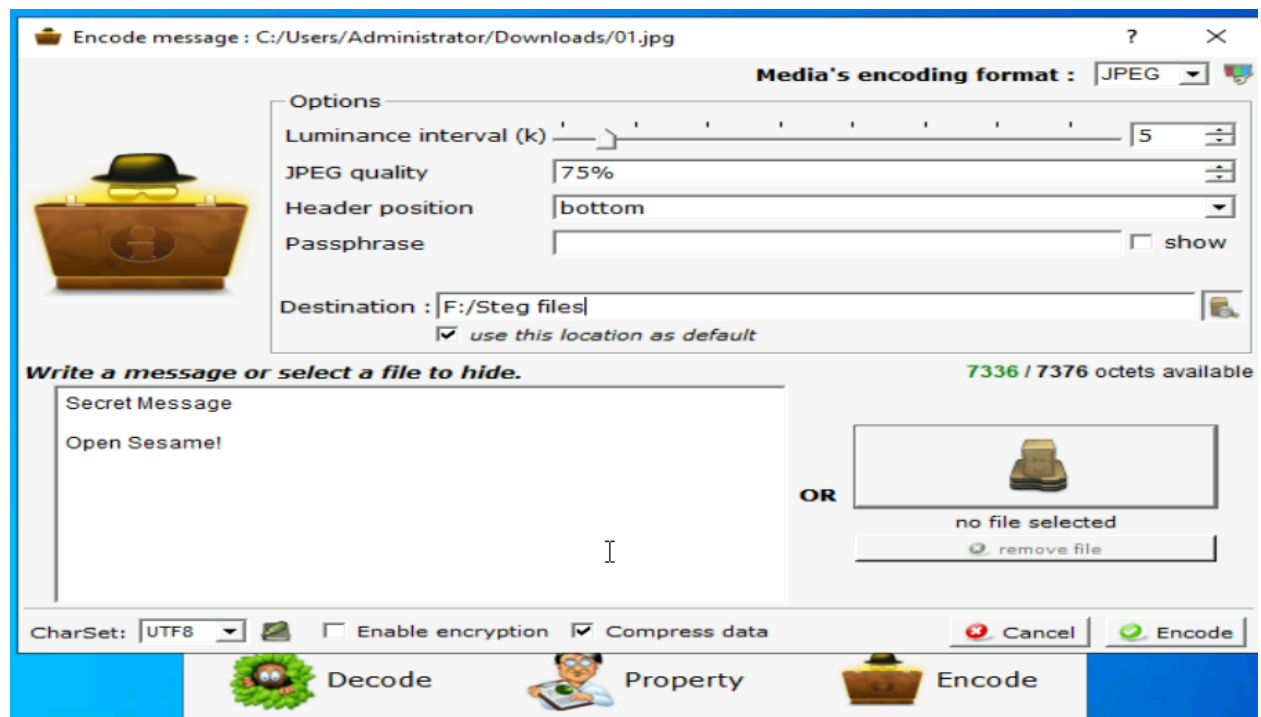


2. Prepare the JPG file for testing

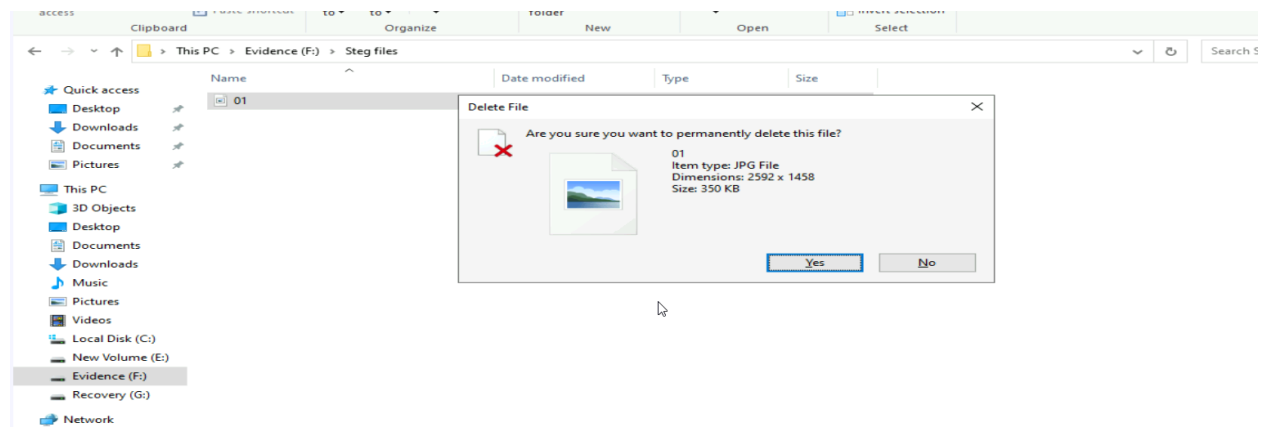
- 2.1. Download and install SilentEye program.
- 2.2. Open SilentEye. Click on the *Open file* button and select the test jpg file(1.jpg).



- 2.3. Then click on *Encode*.
- 2.4. In the new window, remove the default *Passphrase*, set the destination to the newly created evidence volume F drive, and in the bottom right text box type a message to embed in the file.
Then finally click on Encode.

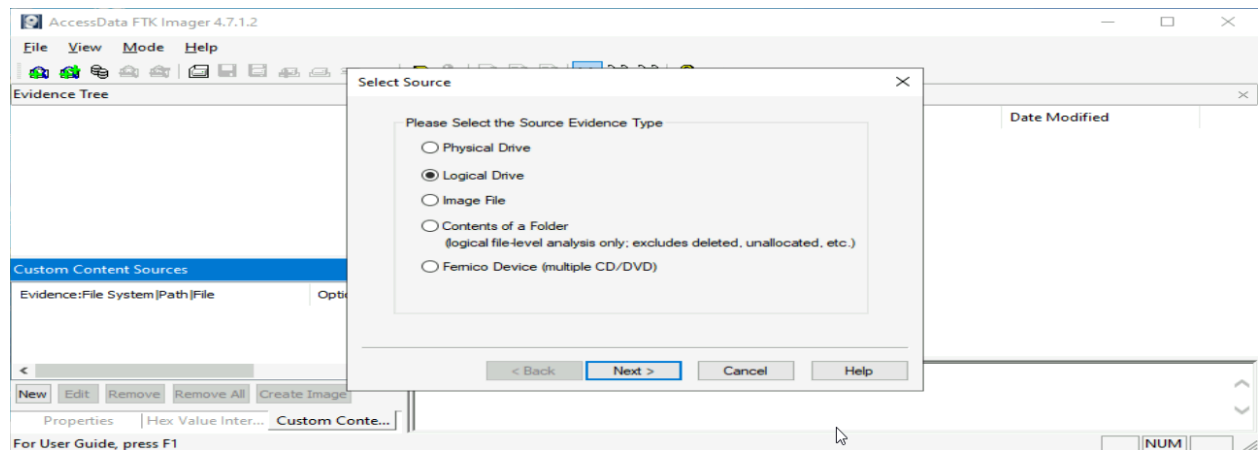


- 2.5. Delete the embedded test file.

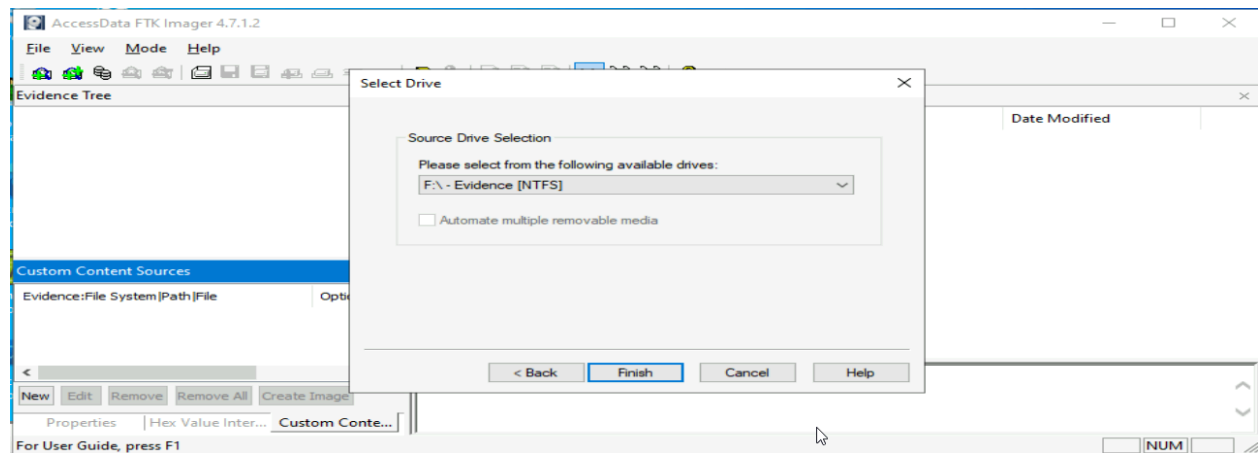


3. Recovery and decoding

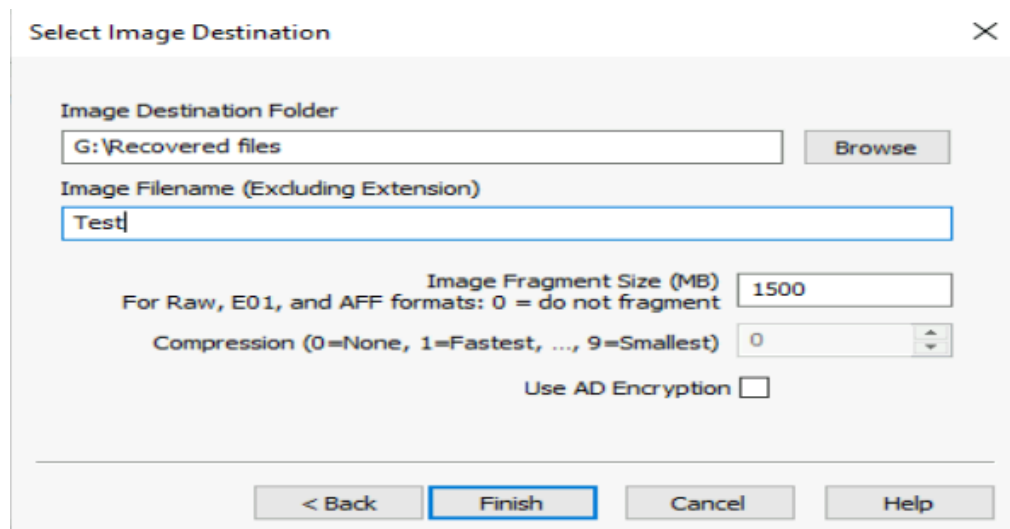
- 3.1. Download, install and run FTKImager.
- 3.2. Click on File, then Create Disk Image.
- 3.3. In the new window, select source as Logical Drive and click on Next.



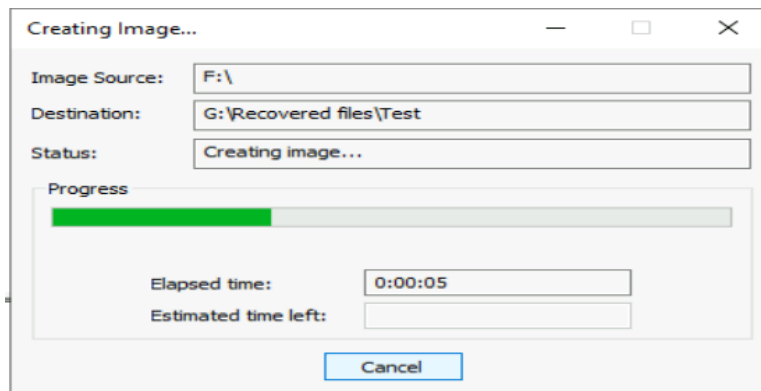
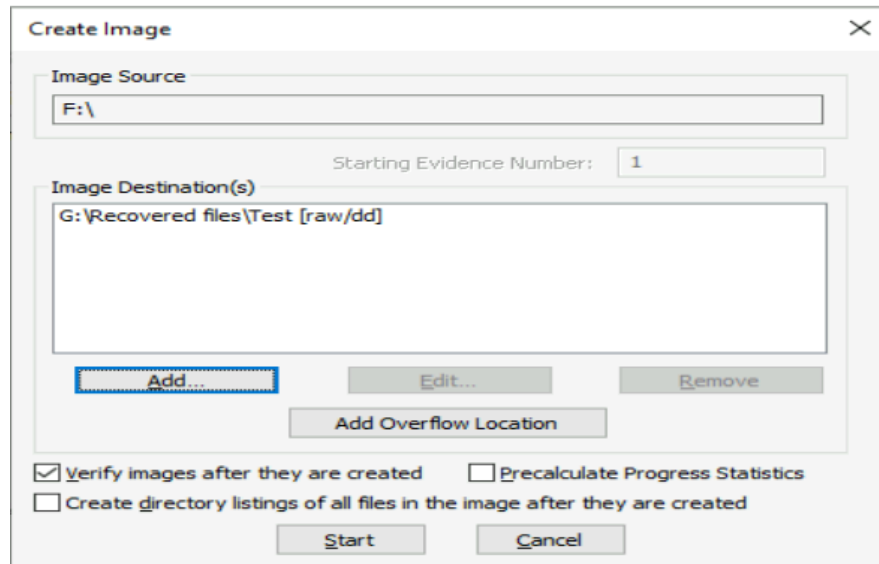
- 3.4. Under Drive selection, select F Drive and click on Finish.



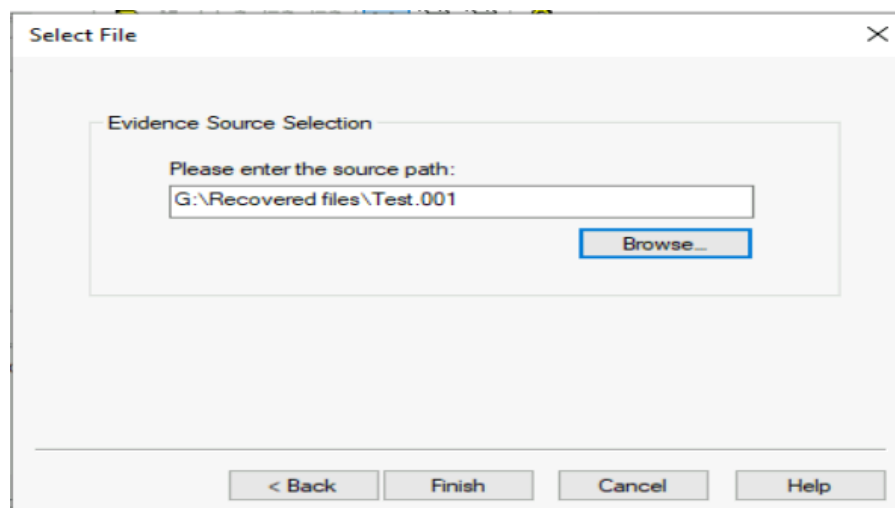
- 3.5. In the next window, select G drive as destination folder, give a filename for the Image and click on Finish



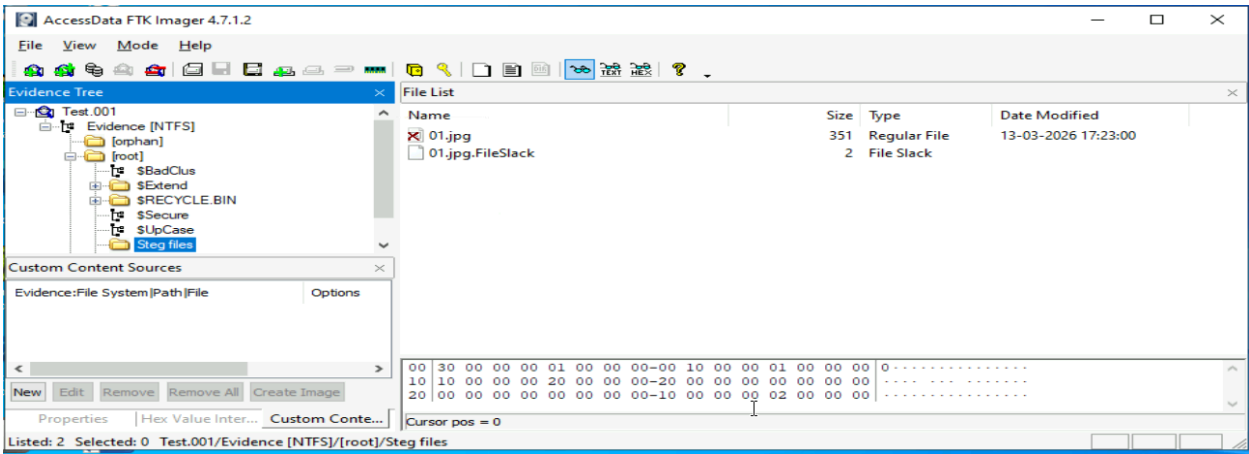
- 3.6. Finally click on Start to create the image files. Wait for the process to complete.



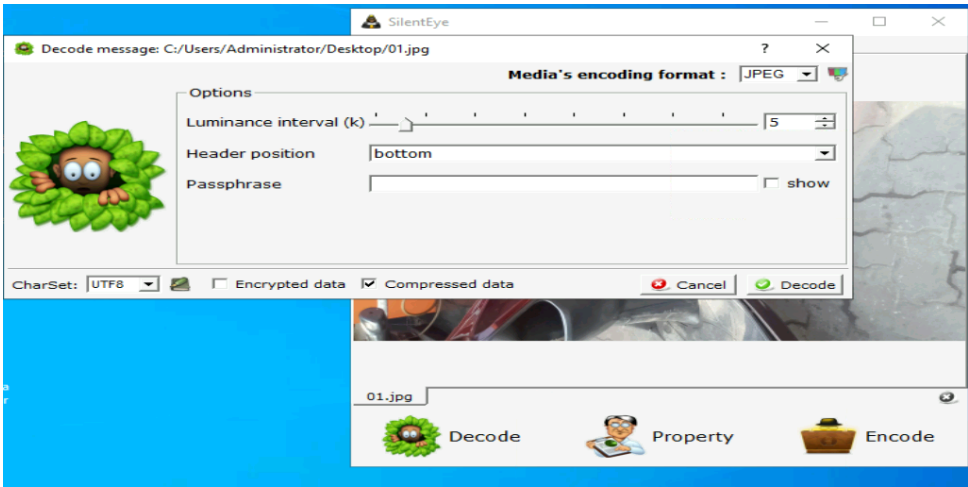
- 3.7. Once the image files are created, click on File, then Add evidence.
- 3.8. For the source path, select the newly created disk image file. Then click on Finish.



- 3.9. Back in the first window, you can see the Evidence tree on the left pane. Expand the tree and navigate to the evidence folder. Test 001 -> Evidence (NTFS) -> [root] -> Steg files. Once the evidence folder is selected, you can see the deleted files on the right window pane.



- 3.10. Right click on the test jpg file (1.jpg) and select Restore to restore the deleted file.
- 3.11. Open SilentEye, select the restored jpg file and click on Decode.
- 3.12. In the new window, remove the passphrase and click on Decode.



3.13. We have successfully restored and decoded the encoded and deleted message.

